

Figures - Research Paper

Explainable Artificial Intelligence for Heart Disease Prediction: A Comparative Analysis of SHAP and LIME Using Random Forest Models

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Figure 1

Proposed Explainable AI Framework

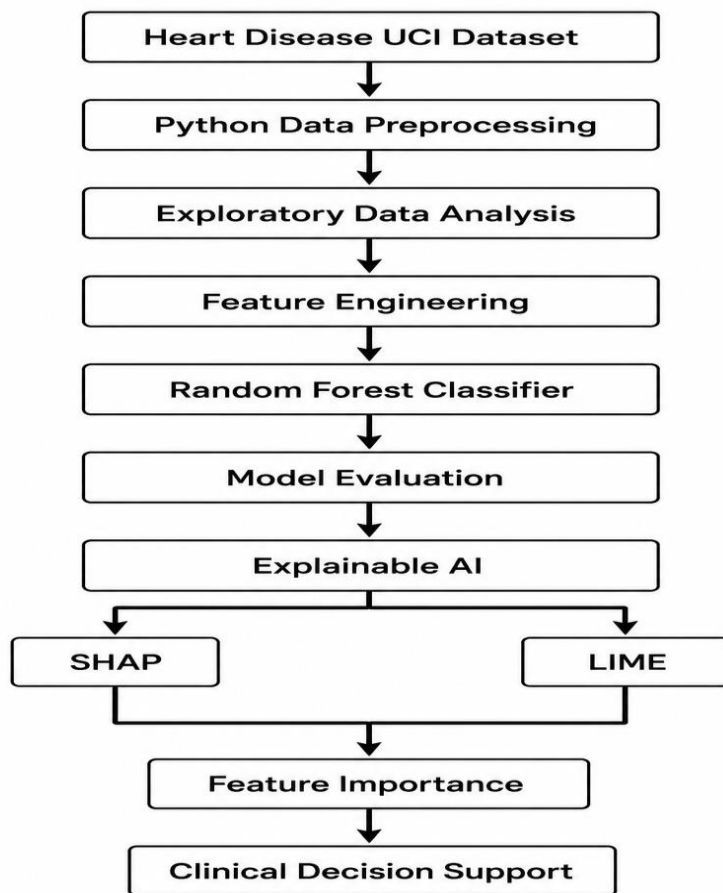


Figure 1. Proposed Explainable AI Framework illustrating the complete workflow from the Heart Disease UCI dataset through data preprocessing, exploratory data analysis, feature engineering, Random Forest classification, Explainable AI techniques (SHAP and LIME), feature importance analysis, and clinical decision support.

Figure 2
Dataset Statistics

Metric	Value
Records	920
Features	16
Features After Encoding	21
Missing Values	Handled
Target Classes	2

Figure 2. Dataset statistics summarizing the characteristics of the Heart Disease UCI dataset, including the number of records, original features, encoded features, handled missing values, and target classes.

Figure 3
Random Forest Feature Importance

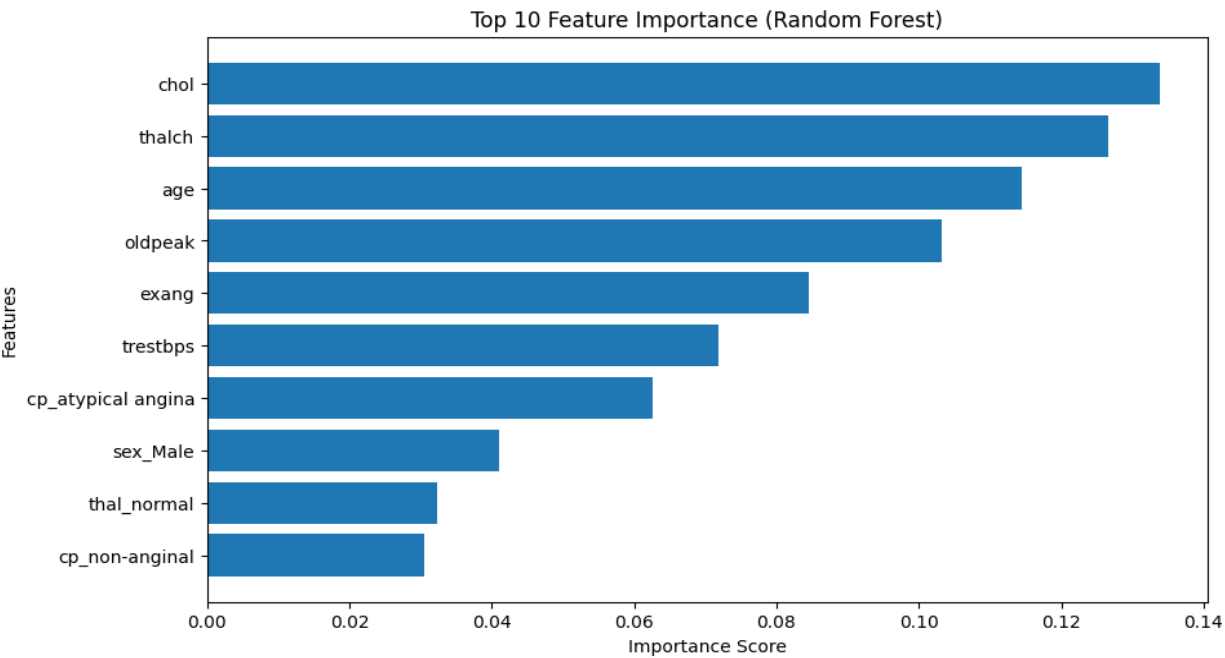


Figure 3. Top ten most important features identified by the Random Forest classifier based on impurity-based feature importance scores.

Figure 4

SHAP Summary Plot

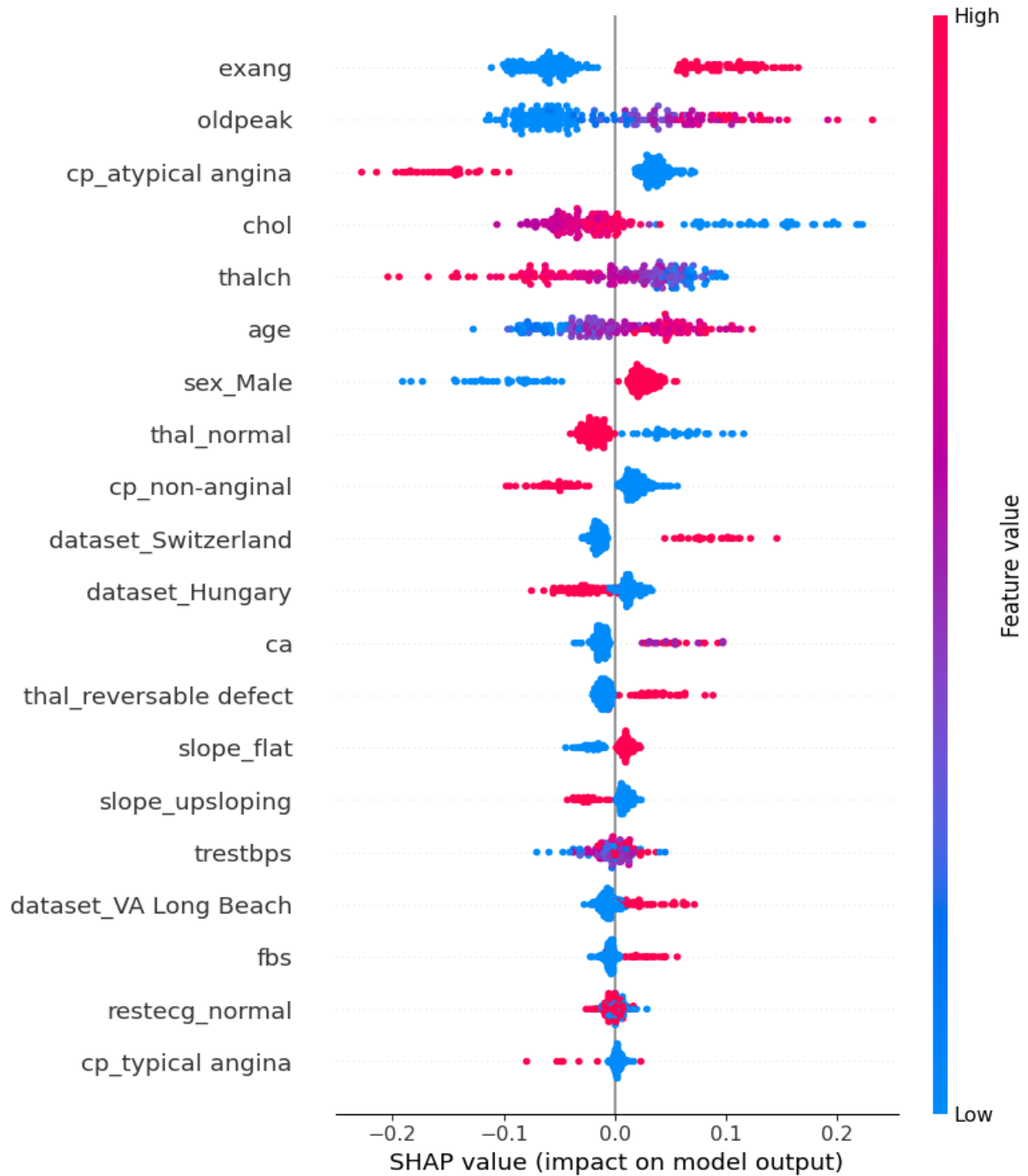


Figure 4. SHAP Summary Plot illustrating the global contribution of each feature to heart disease prediction. Feature importance is represented by the magnitude of SHAP values, while color indicates feature values.

Figure 5

SHAP Feature Importance

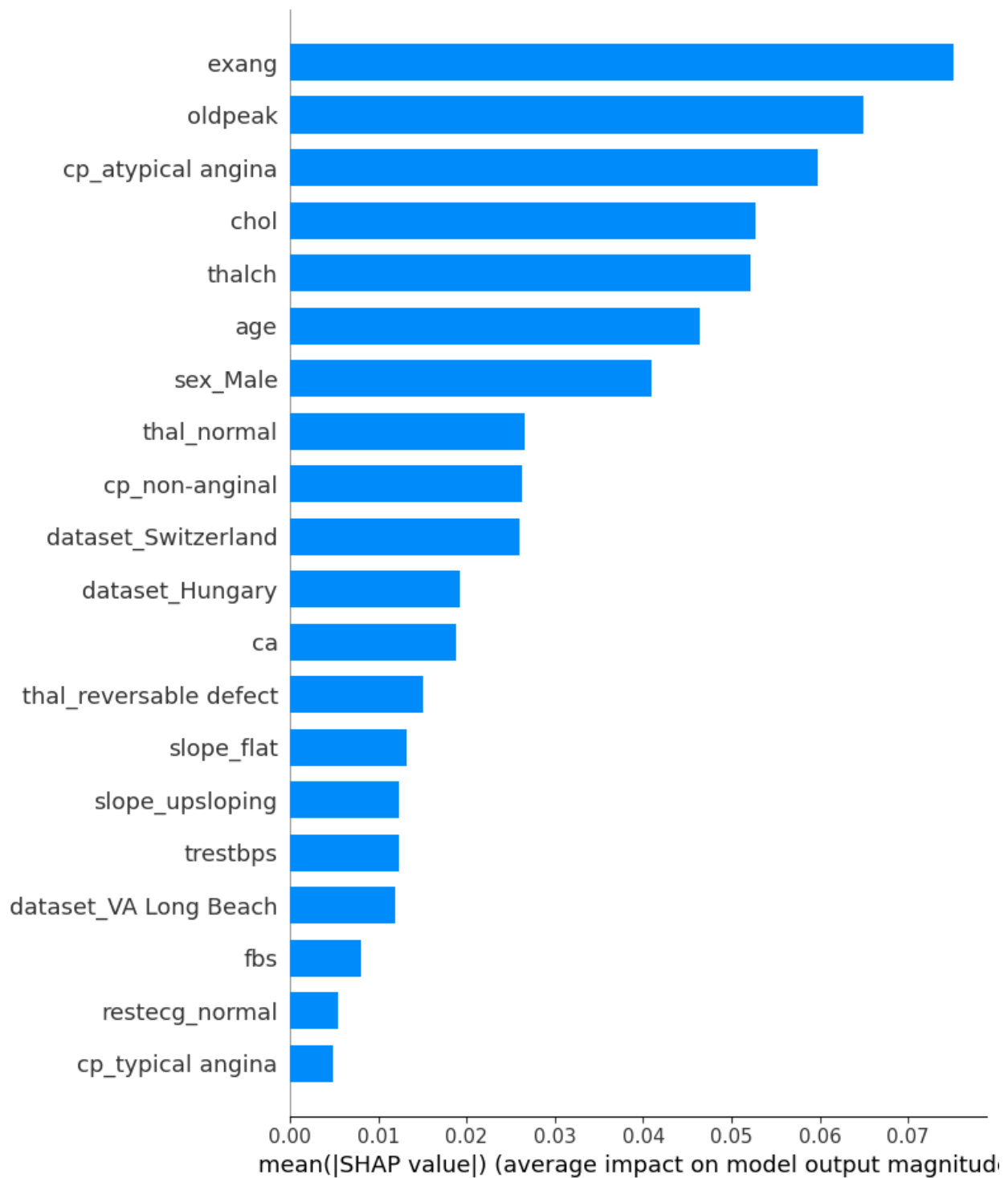
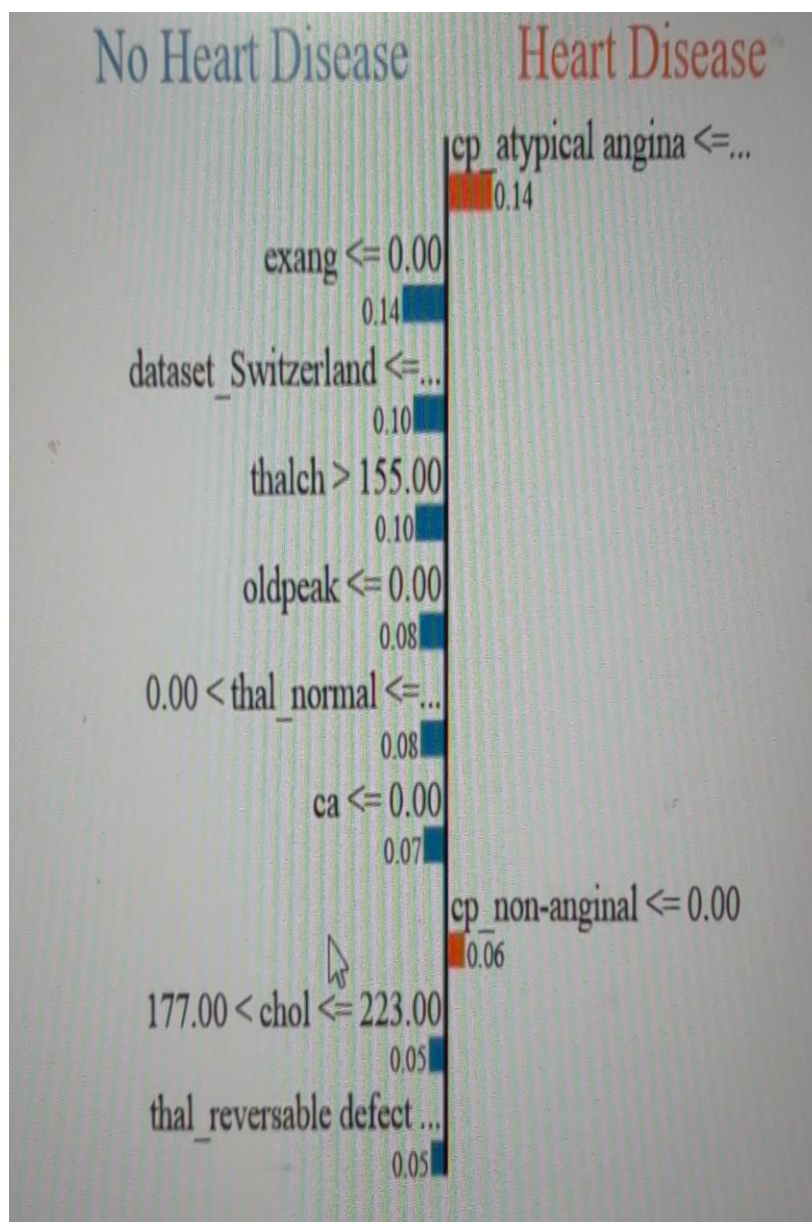
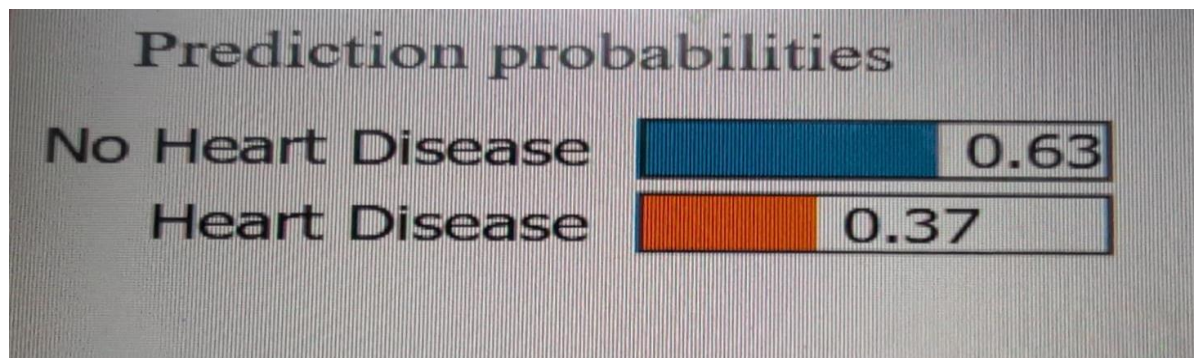
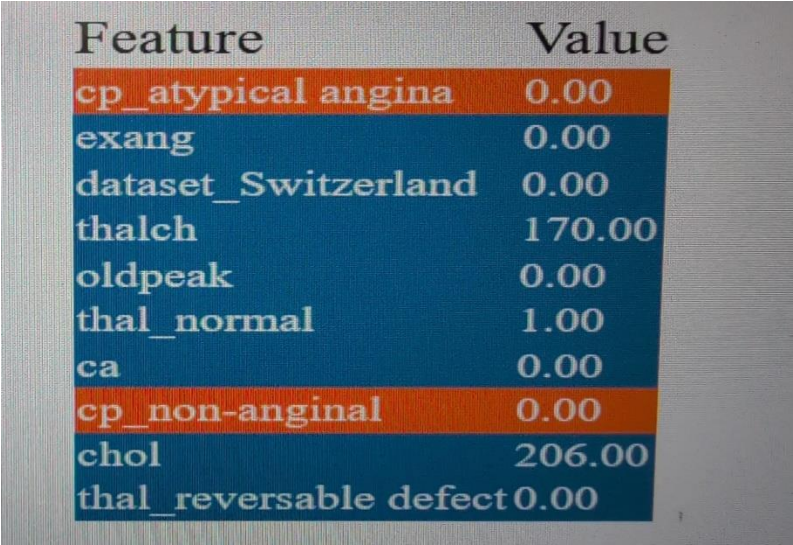


Figure 5. Mean absolute SHAP values showing the global importance of ranking of features influencing the Random Forest model.

Figure 6

LIME Explanation





Feature	Value
cp_atypical angina	0.00
exang	0.00
dataset_Switzerland	0.00
thalch	170.00
oldpeak	0.00
thal_normal	1.00
ca	0.00
cp_non-anginal	0.00
chol	206.00
thal_reversible defect	0.00

Figure 6. Local Interpretable Model-Agnostic Explanation (LIME) demonstrating feature contributions for an individual patient prediction.

Figure 7

Model Performance

Metric	Value
Accuracy	84.24%
Precision	0.84
Recall	0.84
F1-score	0.84
ROC-AUC	0.8363

Figure 7. Performance metrics of the Random Forest classifier, including Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Figure 8
Confusion Matrix

		Predicted	
		Predicted No	Predicted Yes
Actual	Actual No	64	18
	Actual Yes	11	91

Figure 8. Confusion Matrix illustrates the classification results of the Random Forest model on the testing dataset.

Table 1
Dataset Description

Attribute	Description
Dataset	Heart Disease UCI
Patients	920
Features	16
Target	Heart Disease

Table 1. Summary of the Heart Disease UCI dataset, including the number of patients, features, and prediction target.

Table 2

Feature Importance

Rank	Feature	Importance
1	Cholesterol	0.1339
2	Maximum Heart Rate	0.1266
3	Age	0.1145
4	Oldpeak	0.1033
5	Exercise-Induced Angina	0.0845

Table 2. Top five most influential features identified by the Random Forest model.

Table 3

SHAP vs. LIME Comparison

Criteria	SHAP	LIME
Global Explanation	✓	✗
Local Explanation	✓	✓
Computational Cost	High	Low
Consistency	High	Moderate
Interpretability	Excellent	Excellent

Table 3. Comparative analysis of SHAP and LIME based on explainability, computational cost, consistency, and interpretability.

Table 4

Quantitative Evaluation

Metric	Random Forest
Accuracy	84.24%
ROC-AUC	0.8363
Features	21
Samples	920

Table 4. Overall performance evaluation of the Random Forest classifier on the Heart Disease UCI dataset.